Name: BRH-2 Date: 5/12/2026 **Class:** HS Engineering, WTMA

MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details *(A1, A2, A5)*

The problem being addressed takes place at Sky Ranch Middle School in Nevada, specifically within the school library during daily book checkout times. The individuals involved in this problem include the engineering design team (us), Mrs. Hybarger, and the middle school students who use the library. Mrs. Hybarger, the librarian, is a primary stakeholder because she oversees the library environment and has requested a solution that increases student engagement while they wait in line. The engineering team is responsible for designing, building, and testing the “This or That” machine based on her needs and feedback. Middle school students are also primary stakeholders because they are the main users of the system and are directly affected by the problem the design is intended to solve.

This problem occurs when students are waiting in line to check out or return books in the library and have no structured activity to keep them engaged. During these moments, students often experience boredom or disengagement, especially when lines are long or slow-moving. This is particularly noticeable among quieter or more reserved students who may not participate in larger group activities or initiate conversation easily. As a result, there is a need for a low-pressure, interactive solution that allows all students to participate equally while waiting.

The issue happens on a daily basis during school hours whenever the library is open, since students visit the library regularly throughout the week for reading, research, and book checkout. Because of this frequent usage, the lack of engagement opportunities becomes a recurring problem rather than a one-time issue. The cause of the problem is not a lack of interest in the library itself, but rather the idle time students experience while waiting in line without any structured interaction or activity.

Research shows that student engagement in school environments plays an important role in participation and social interaction. According to the National Center for Education Statistics, school libraries are present in over 95% of public schools in the United States, making them a common space where students spend time outside of classrooms (NCES, 2024). Additionally, studies in educational research suggest that interactive and peer-based activities can improve communication skills and student engagement in school settings (National Library of Medicine, 2023). This supports the need for a solution that encourages interaction even during short waiting periods.

In order to address this issue, a “This or That” interactive machine is being developed to give students a quick and simple way to participate in decision-based prompts while waiting in line. This solution aims to reduce boredom, increase social interaction, and improve the overall library experience for students and staff.

Works Cited

“News from the National Library of Medicine - 2023.” *Www.nlm.nih.gqov*, www.nlm.nih.gov/news/2023.html.

“The NCES Fast Facts Tool Provides Quick Answers to Many Education Questions (National Center for Education Statistics).” *Nces.ed.gov*, 2024, nces.ed.gov/fastfacts/display.asp?id=42.

Stakeholder Concerns and Connections *(A3, A4)*

Mrs. Hybarger is the librarian for whom this design is being created, making her one of the primary stakeholders. She is directly impacted by the final product because the machine will be installed in the library and used daily. Her main concerns include ensuring the device is safe for students, durable enough for constant use, easy to manage during busy library periods, and engaging enough to improve student behavior and participation while waiting in line.

Middle school students at Sky Ranch Middle School are another key primary stakeholder group because they are the main users of the “This or That” machine. These students are directly affected by boredom and lack of engagement while waiting in line. Their concerns include having something fun, quick, and easy to interact with that does not create pressure or embarrassment. The design must also be simple enough for all students, including quieter or more reserved individuals, to participate comfortably.

The engineering students designing and building the machine are also primary stakeholders. Their concerns include creating a working system that accurately detects inputs, displays results correctly, and remains reliable over repeated use. They are also responsible for meeting all design requirements, ensuring safety, and successfully integrating both the physical structure and electronic components.

Teachers and school staff are additional primary stakeholders because the machine will operate in a shared educational environment. Their concerns include minimizing distractions, ensuring student safety, and making sure the device contributes positively to the learning environment rather than disrupting it. They also want the machine to support positive behavior and smooth transitions during busy library times.

Field experts and mentors, including Mrs. Hybarger and engineering instructors, are also stakeholders because they guide the design process through feedback and evaluation. Their concerns focus on ensuring the design meets engineering standards, functions correctly, and can be improved through testing and iteration.

Secondary stakeholders include parents and other school community members who may not directly use the machine but are still affected by its impact. Parents may benefit from increased student engagement and improved social interaction at school. Additionally, the school environment as a whole is affected because a successful design could improve library culture by encouraging communication and reducing idle boredom. There are also minor environmental considerations, as using reusable electronic components instead of paper-based activities can reduce waste over time.

Problem Importance *(A6)*

This problem is important because many middle school students experience boredom or social discomfort while waiting in lines at school. Interactive activities can help students feel more engaged and connected with others. According to the National Center for Education Statistics, over 95% of schools in the United States have libraries that support student learning and social interaction. Libraries are important shared spaces where students spend time outside of classrooms.

Research has also shown that interactive activities and student engagement can improve social connection and participation among students. A study published by the National Library of Medicine found that interactive group activities can help students build confidence and communication skills (National Library of Medicine, 2021).

This problem should be solved now because students use the library every day, meaning the issue happens frequently. By creating a “This or That” machine, students can stay engaged while waiting in line, communicate with peers, and make the library environment more welcoming and interactive.

Works Cited

"Confidence-building in a collaborative, multidisciplinary mock page activity for fourth-year medical students" *Nih.gov*, 2024, pmc.ncbi.nlm.nih.gov/articles/PMC9897763/ Accessed 15 May 2026.

National Center for Education Statistics. "Concentration of Public School Students Eligible for Free or Reduced-Price Lunch." *Nces.ed.gov*, May 2022, nces.ed.gov/programs/coe/indicator/clb/free-or-reduced-price-lunch.